

Data Analyst Project: Patient Records Healthcare Sector

Overview of Project

This report analyses patient records from a hospital dataset, examining the relationships across five tables patient (demographics lifestyle factors), diagnoses, medications, lab results and hospital outcomes. Despite the volume of data collected, identifying high-risk patients remains a significant challenge for healthcare providers.

The analysis was conducted using SQL (PostgreSQL) for data extraction, cleaning, and querying across all tables. The cleaned dataset was then exported to Power BI for visualisation and reporting.

The dataset was sourced from Kaggle and covers over 100,000 patients aged 19 to 95, and 15 disease categories.

Problem Statement

Healthcare providers face increasing pressure to reduce costs while maintaining high-quality patient care. These analyses seek to address the following central questions:

- Who are the patients and what are their key demographics?
- Which lifestyle factors are associated with poor outcomes such as readmission and in-hospital death?
- Does medication adherence help reduce adverse outcomes?
- How can healthcare costs be reduced without compromising patient care?
- What department has the most patients?

Project Goal

The goal of this project is to analyse patient data to determine whether age influences medical conditions, and to identify which patient profiles carry the highest risk of readmission, ICU admission, elevated costs, and in-hospital death. The insights generated are intended to support data-driven decisions that improve both operational efficiency and patient care outcomes.

Data Cleaning

- Identified and handled null values, duplicate records, and statistical outliers across all five tables.
- Replaced missing values in the days-to-readmission column with 0 to maintain data integrity and ensure consistency during analysis.
- Replaced null entries in the secondary diagnoses and secondary ICD-10 columns with none.

Analysis

Analyses were performed across five tables using JOIN queries connected through patient ID, enabling access to better results output and getting the bigger picture

Key Findings

- Lifestyle factors (including smoking, alcohol use, and exercise level) did not produce differences in length of stay, total charges, or readmission rates. The variation across lifestyle groups was too small to be meaningful.
- Lifestyle factors did not appear to directly influence primary diagnoses. Patients classified as having a healthy lifestyle (non-smokers who exercised and abstained from or did light alcohol) recorded the highest number of diagnosed conditions. This suggests that healthier individuals may engage more with healthcare services, resulting in higher detection rates.
- Contrary to expectations, patients with high medication adherence were readmitted at a rate of 44.6%, compared to 18.4% among low-adherence patients. This does not indicate that medication is not working but rather, it suggests that disease severity is the primary driver of readmission. Sicker patients tend to be more diligent with medication yet remain at higher risk due to the complexity of their conditions.
- The most common primary diagnoses are hypertension, chronic kidney disease (CKD), obesity, and hyperlipidaemia. These are long term manageable conditions. and these diagnoses suggests that many hospitalisations may be preventable through stronger outpatient and preventive care.
- Chronic kidney disease is the leading cause of in-hospital death, followed by obesity and hypertension.
- Costs remain broadly consistent regardless of individual risk factors. No single factor (including ICU admission, insurance type, primary diagnosis, or readmission) drives a significantly higher average cost compared to others. This indicates that the hospital cost is primarily a volume problem rather than a severity problem.
- Type 1 diabetes patients recorded the longest average length of hospital stay.
- Uninsured patients had the lowest average costs, compared to those insured. However, deeper analysis revealed that uninsured patients are less likely to complete their full course of treatment, likely due to financial constraint, which increases their probability of readmission. TRICARE (military insurance) recorded the highest average charges, reflecting the more intensive treatment plans.
- Most primary diagnoses are managed collaboratively across multiple departments but internal medicine has twice the patients as the other department .
- Elderly patients exhibit a death rate more than 10 times higher than any other age group, highlighting the need to focus on elderly care.
- Of the 245 patients admitted to the ICU, 136 died. This strongly suggests that patients are reaching the ICU at the stage of deterioration, when intervention is least effective. Additionally, all patients who died in hospital were not readmitted, as their outcomes were fatal. All patients who survived and were readmitted within 30 days ultimately lived.

Recommendations

1. **Establish a chronic disease outpatient programme.** Given that the most common diagnoses are hypertension, CKD, obesity, and hyperlipidaemia are all long-term manageable conditions, the hospital should invest in structured outpatient monitoring and patient education programmes. Intercepting these conditions before they escalate to hospital admission would reduce patient volume, lower overall costs, and improve mortality outcomes.
2. **Implement an early deterioration warning system.** With a 55% ICU mortality rate (136 of 245 patients), the data indicates that patients are being escalated to intensive care too late. An early warning scoring system would enable clinical teams to identify at-risk patients earlier, intervene before conditions become critical, and either treat effectively or safely discharge those who do not require intensive care.
3. The mortality rate among elderly patients is more than 10 times that of other age groups. A dedicated clinical pathway for patients aged 60 and above should include extended observation periods before discharge, mandatory follow-up appointments within 48 hours of discharge, and assigned care giver for those managing any of the identified chronic conditions.
4. **Financial assistance for uninsured patients.** Uninsured patients leave before completing treatment due to cost concerns, which increases their likelihood of readmission and generates higher long-term costs for the hospital. Establishing a financial assistance programme or flexible payment plan would help uninsured patients complete their treatment, reducing readmissions and improving outcomes.
5. Since medication adherence did not predict readmission in the expected direction, the hospital should redirect risk efforts towards the Charlson Index and diagnosis severity at admission as primary readmission predictors. Resources currently allocated to compliance monitoring could be redirected towards disease management programmes and clinical support.
6. **Promote generic medication that are legal:** whether medications are generic or branded, they contain the same ingredients. A review of prescribing patterns could identify opportunities to switch to cost-equivalent generic alternatives, reducing medication costs for both the hospital and patients without affecting affect good healthcare

Preventive outpatient care, earlier intervention, and better post-discharge follow-up are the most effective levers for cost reduction without compromising care quality.

NOTE: This analysis is based on a synthetic dataset sourced from Kaggle, in which values were randomly distributed. All key findings reflect patterns observed within this dataset and should not be interpreted as conclusions drawn from real-world clinical data.

